

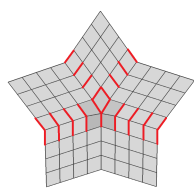
Patching Non-Uniform Extraordinary Points with Sharp Features

Yi-Fei Feng
UCAS
Beijing, China
fengyifei15@mails.ucas.ac.cn

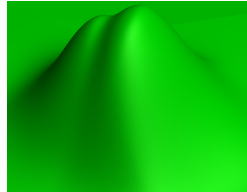
Xin Li
USTC
Hefei, China
lixustc@ustc.edu.cn

Chun-Ming Yuan
AMSS, CAS
Beijing, China
cm yuan@mmrc.iss.ac.cn

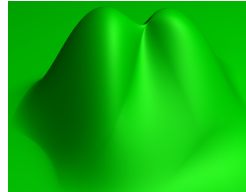
Li-Yong Shen
UCAS
Beijing, China
lyshen@ucas.ac.cn



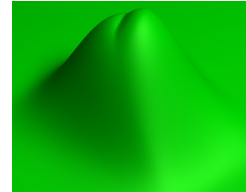
(a) Given control grid



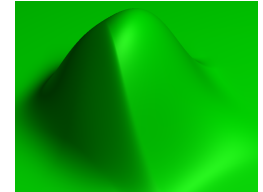
(b) result produced by [Cashman et al. 2009]



(c) result produced by [Scott et al. 2013]



(d) result produced by [Sederberg et al. 1998]



(e) result of our method

Figure 1: The surfaces generated from a control grid with non-uniform parameterization, where the knot intervals for the red edges are six and the other knots are ones.

ABSTRACT

Extending the non-uniform rational B-spline (NURBS) representation to arbitrary topology is one of the most important steps to define the iso-geometric analysis (IGA) suitable geometry. The approach needs to be NURBS-compatible and can handle non-uniform knot intervals. To achieve this goal, we present a novel patching solution which define one Bézier patch for each non-zero knot interval control grid face. The construction can reproduce the bi-cubic NURBS in the regular face and define bi-quintic Bézier patches for irregular faces. The method can also support non-uniform sharp features along the extraordinary points. Experimental results demonstrate that the new surfaces can improve the surface quality for non-uniform parameterization.

KEYWORDS

NURBS, capping, non-uniform, G^1 continuous, sharp feature

ACM Reference Format:

Yi-Fei Feng, Xin Li, Chun-Ming Yuan, and Li-Yong Shen. 2021. Patching Non-Uniform Extraordinary Points with Sharp Features. In *SIGGRAPH Asia 2021 Posters (SA '21)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3476124.3488626>

1 INTRODUCTION

Defining the smooth surfaces from an arbitrary topological control grid with non-uniform knot intervals is a foundation problem both for computer-aided design (CAD) and iso-geometric analysis (IGA).

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '21, December 14–17, 2021, Tokyo, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8687-6/21/12.

<https://doi.org/10.1145/3476124.3488626>

Several surface representations have been discovered in recent years using non-uniform subdivision [Cashman et al. 2009; Kovacs et al. 2015; Li et al. 2016; Sederberg et al. 1998] and patching [Peters 2000; Scott et al. 2013].

We use the term of knot interval ratio of an extraordinary point (EP) to mean the largest knot interval of an edge adjacent to the extraordinary point divided by the smallest. One problem for which most subdivisions and patching methods alike fail to work well is whose EPs edges have widely varying knot ratios, such as the surfaces in Figure 1 and the results in [Li et al. 2016]. The problem of first observed in [Sederberg et al. 1998] and several heuristic approaches have been researched in order to improve the surface quality in [Cashman et al. 2009; Kovacs et al. 2015]. Li et al. created acceptable subdivision surface for any reasonable choice of non-uniform knot intervals in [Li et al. 2016]. On the other hand, sharp features on surfaces play an important role in geometric modelling. To our best knowledge, the complex non-uniform sharp features can be created only with non-uniform subdivision [Tian et al. 2020].

However, subdivision methods are not backward compatible with NURBS because near EPs, these methods produce an infinite sequence of bi-cubic patches, of which NURBS can only import a finite truncation. Therefore, it is still open to define a smooth surface from an arbitrary topological control grid with non-uniform parameterization in the NURBS-compatible way and to generate complex sharp features on such a representation. This motivates us to propose a NURBS-compatible approach for constructing smooth surfaces of arbitrary control grid with large knot interval ratios. The framework is patch based so it is backward (and forward) compatible with NURBS. Our basis functions are guaranteed to be global G^1 smooth, which can be used in downstream applications directly. Moreover, the method can also support non-uniform sharp features passing extraordinary points, which is significant for geometric modeling in practical applications.

2 METHODOLOGY

We are given an arbitrary topological quadrilateral control grid, which edges of the grid are assigned with a non-negative knot intervals. The only restriction for the knot intervals is that the intervals on the opposite edges of each face must be same. Some edges of the control grid can be marked with sharp edges. And then we can define a set of basis functions, which is C^2 continuity on regular edges, G^1 continuity on spoke edges and C^0 along the sharp edges. Comparing the exists patching approaches, our method has three main contributions. First our method has no restrictions on the EPs of the present control grid, which faces can have more than one EPs. Second our method can work well on non-uniform cases of the extension of bi-cubic B-spline. Third our method can easily produce the sharp edges as needed.

The non-uniform knot intervals will lead some new challenges. First, the knot intervals around the EPs are assymetric which makes the definition of the face points to be ambiguous. Second, the connecting functions for the G^1 continuous are not well defined for the non-uniform knot intervals. Third, no tools can handle both G^1 -continuous and sharp features. In order to solve these problems, our main ideas are introduced in basic framework of our construction as below.

Initial Basis functions construction. In order to improve the surface quality as well as the convergent rate for the isogeometric analysis, we add n auxiliary points to replace each valence n EP. And then we perform the Bézier extraction to get C^0 basis functions. Each face corresponds a bi-cubic Bézier patch. With the auxiliary points, we can avoid the ambiguous knot intervals. The auxiliary points are optimized to improve the surface quality.

Define the connecting functions. The G^1 continuity between two patches depends on the first cross-derivatives only, which was characterized by the connecting functions. As the connecting functions are expected to adapt to the non-uniformity of the knot, we divide the angle of the tangent plane on EP according to the knot intervals, then define the connecting functions as the function of the angles which can define the smooth transaction between G^1 and C^1 continuous.

Optimized the G^1 basis functions. We use bi-quintic Bézier patch to ensure the G^1 continuity in irregular faces. In order to avoid global optimization of basis functions, we solve G^1 continuity constraints in two steps: First we optimize the control points around EPs, which are affected by different constraints. Second we solve the constraints in each edge and make the spoke edge close to cubic curve. After we get the G^1 solutions with certain free variables, we optimize these variables to make the entire surface smoother.

3 RESULT

Figure 1(a) shows a non-uniform control mesh with a val-5 extraordinary point. According to Figure 1, other methods can not keep unimodal shape except our method 1(e) when generating G^1 continuous surface near a non-uniform EP.

Our method can easily create creases along arbitrary mesh edges. Figure 2(left) define a non-uniform control grid with val-5 EP and sharp edges. All sharp edges are pass through EP and the two blue sharp edges are continue while purple sharp edge ends at the EP.

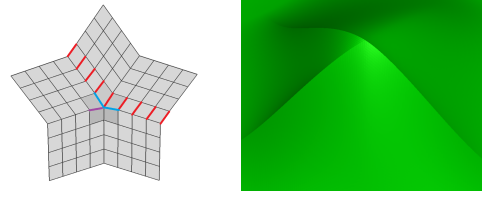


Figure 2: Create creases through EP

Figure 2(right) shows the sharp feature we create near EP. For the ring model, we set red knot interval to two in the non-uniform grid in Figure 3(left), and blue edges are corresponding to a continue crease (see Figure 3(right)).

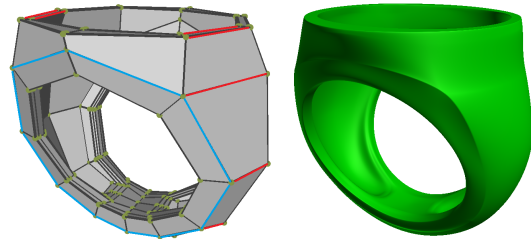


Figure 3: Crease on ring model

4 CONCLUSION

We propose a non-uniform spline function on arbitrary control mesh with global G^1 continuity. The surface quality remains good in large knot interval ratios case. And it can easily create creases along control mesh edges even passing EPs.

ACKNOWLEDGMENTS

This work was partially supported by the Beijing Natural Science Foundation (No. Z190004), the National Natural Science Foundation of China (No. 61872332) and the University of Chinese Academy of Sciences.

REFERENCES

- T. J. Cashman, U. H. Augsdörfer, N. A. Dodgson, and Malcolm A. Sabin. 2009. NURBS with Extraordinary Points: High-degree, Non-uniform, Rational Subdivision Schemes. *ACM Transactions on Graphics* 28, 3 (2009), 1–9.
- Denis Kovacs, Justin Bisceglia, and Denis Zorin. 2015. Dyadic t-mesh subdivision. *ACM Transactions on Graphics (TOG)* 34, 4 (2015), 143.
- Xin Li, G Thomas Finnigan, and Thomas W Sederberg. 2016. G^1 non-uniform Catmull-Clark surfaces. *ACM Transactions on Graphics (TOG)* 35, 4 (2016), 1–8.
- Jörg Peters. 2000. Patching catmull-clark meshes. In *SIGGRAPH'00: Proceedings of the 27th annual conference on Computer graphics and interactive techniques*. 255–258.
- Michael A Scott, Robert N Simpson, John A Evans, Scott Lipton, Stephane PA Bordas, Thomas JR Hughes, and Thomas W Sederberg. 2013. Isogeometric boundary element analysis using unstructured T-splines. *Computer Methods in Applied Mechanics and Engineering* 254 (2013), 197–221.
- T.W. Sederberg, J. Zheng, D. Sewell, and M. Sabin. 1998. Non-uniform recursive subdivision surfaces. In *SIGGRAPH'98: Proceedings of the 25th annual conference on Computer graphics and interactive techniques*. ACM Press/Addison-Wesley Publishing Co., 387–394.
- Yufeng Tian, Xin Li, and Falai Chen. 2020. Non-Uniform Subdivision Surfaces with Sharp Features. *Computer Graphics Forum* 39, 6 (2020), 232–242.